Extraction Prompt v3a

Eliminating Source-Value Leakage

ArcMemo Concept Extraction — Fast Iteration Results

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The Leakage Problem

In the m3 experiment, **76% of extracted math concepts** (634/837) contained source-specific numeric values in their `cues` and `implementation` fields.

Root cause

The extraction prompt asked for *"how this concept was applied in this specific solution"* — producing entries like:

- `"compute 160mi / 5hr = 32mph"`
- `"base CM = 2, height = 8 → area = 8"`
- `"center at (4,4) for a 7×7 grid"`

These values **leak into hints for different target problems**, causing misleading guidance.

Leaky Example 1: Speed (source: cmath_6446)

```yaml
- concept: Speed Unit Interpretation
  kind: definition
  description: >
    speed is distance per unit time; with distance in miles
    and time in hours, the speed unit is miles per hour (mph)
  cues:
    - "distance measured in miles"
    - "time measured in hours"
  implementation:
    - "compute 160 miles / 5 hours = 32 mph"       # LEAKED!
```

When selected as a hint for a *different* problem (e.g., trains traveling 300 km in 4 hours), the solver sees **"160 miles / 5 hours = 32 mph"** and incorporates irrelevant numbers.

Leaky Example 2: Geometry (source: cmath_5659)

```yaml
- concept: Triangle Area Formula
  kind: theorem
  cues:
    - "used to compute ACM with base CM and height
       distance from A to the line through CM (x = 8)"   # LEAKED!
    - "used to compute ACN with base CN and height
       distance from A to the line through CN (y = 0)"   # LEAKED!
  implementation:
    - "ACM: base CM = 2, height = 8 → area = 8"         # LEAKED!
    - "ACN: base CN = 4, height = 4 → area = 8"         # LEAKED!
```

Cues reference specific points (A, C, M, N) and coordinates. Implementation has hardcoded numeric answers.

Meaningless for any other problem.

Leaky Example 3: Grid Coloring (source: cmath_2171)

```yaml
- concept: Center-based Relative Coordinate System
  kind: definition
  description: >
    Use coordinates relative to the center of the grid
    to simplify symmetry analysis.
  cues:
    - "center is at (4,4) for a 7×7 grid"                # LEAKED!
    - "dx = i - 4, dy = j - 4 with
       dx, dy in {-3, -2, -1, 0, 1, 2, 3}"               # LEAKED!
  implementation:
    - "classify a point by (|dx|, |dy|) up to order
       to identify its orbit under D4"
```

If this hint is given for a **5x5 grid** or a **hexagonal grid**, the values `(4,4)` and `7×7` are actively misleading.

The Fix: v3a Prompt

v3a Anti-Leakage Instructions

Added to the Stage 2 extraction prompt:

```
- CRITICAL: No problem-specific values. Strip ALL concrete
  numbers, coordinates, coefficients, and computed answers
  from cues and implementation. Write abstract patterns only.

    BAD cue: "center is at (4,4) for a 7×7 grid"
    GOOD cue: "problem has a grid with a natural center point"

    BAD implementation: "compute 160 miles / 5 hours = 32 mph"
    GOOD implementation: "compute speed = distance / time"

    BAD implementation: "ACM: base CM = 2, height = 8 → area = 8"
    GOOD implementation: "compute triangle area = 1/2 x base x height
      using perpendicular distance"
```

Also changed `implementation` field from *"how this concept was applied in this specific solution"* to *"the general procedure (abstract, not specific)"*.

Clean v3a: Speed (same source: cmath_6446)

```yaml
- concept: Average Speed Calculation
  kind: technique
  parameters:
    - name: total distance
      typing: number
      description: the total path length covered
    - name: total time
      typing: number
      description: the total duration of the travel
  description: >
    Calculate average speed by dividing total distance
    traveled by total time elapsed.
  cues:
    - "problem involves motion or travel"
    - "asks for average speed rather than instantaneous speed"
  implementation:
    - "divide total path length by total time duration"
```

No "160 miles", no "5 hours", no "32 mph". **Fully abstract and reusable.**

Clean v3a: Geometry (same source: cmath_5659)

```yaml
- concept: Shoelace Formula
  kind: technique
  parameters:
    - name: vertices
      typing: list[tuple[float, float]]
      description: ordered list of polygon vertex coordinates
  description: >
    Calculate the area of a polygon given the coordinates
    of its vertices in order.
  cues:
    - "polygon vertices are given as coordinates"
    - "area calculation is required"
  implementation:
    - "list vertices in cyclic order"
    - "compute sum of cross-products of adjacent coordinates"
    - "take absolute value and divide by 2"
```

No hardcoded points. No specific areas. **Just the general procedure.**

Clean v3a: Grid Coloring (same source: cmath_2171)

```yaml
- concept: Invariant Colorings via Orbits
  kind: technique
  parameters:
    - name: number of colors
      typing: int
    - name: domain set
      typing: str
      description: set of elements being colored
  description: >
    Count colorings invariant under a group action by
    assigning colors to orbits of the group action.
  cues:
    - "problem asks for colorings invariant under a symmetry group"
    - "elements of the domain partition into orbits"
  implementation:
    - "identify the orbits of the domain under the group action"
    - "total colorings = (number of colors)^(number of orbits)"
```

No `(4,4)`, no `7×7`. **Works for any symmetry group on any domain.**

Code Domain (LiveCodeBench)

v3a Code Concepts

Code domain had minimal leakage (1/168) but needed two additional fixes:

1. **Anti-backtick instruction** — backticks in YAML break parsing
2. **Abstract implementation** — same anti-leakage rules as math

Example: Digit DP

```yaml
- concept: Digit DP
  kind: algorithm
  description: >
    DP approach to count numbers satisfying digit-based
    properties within a range.
  cues:
    - "Problem asks to count integers in a range with digit properties"
    - "Constraints involve divisibility or digit counts"
    - "State depends on position and accumulated properties"
  implementation:
    - "iterate over digit positions, track tight bound and property state"
    - "use memoization on (position, tight, accumulated_state)"
```

v3a Code Concepts (cont.)

Prefix Sums

```yaml
- concept: Prefix Sums
  kind: technique
  description: >
    Precomputing cumulative sums to allow O(1) range
    sum queries on static arrays.
  cues:
    - "Problem requires frequent sum queries over subarrays"
    - "The array is static with no updates"
    - "Range sums needed for multiple different intervals"
  implementation:
    - "Construct array where each index stores cumulative sum
       from start. Calculate subarray sums by subtracting
       prefix at start from prefix at end."
    - "Answer range query [L, R] as count(R) - count(L-1)"
```

Procedural details matter for code — the solver needs to know *how* to apply the algorithm, not just its name.

Results

Leakage Elimination

Version	Math Leakage	LCB Leakage
v1 (original)	*76%* (634/837)	1/168
v3a (anti-leakage)	0% (0/53)	0%

v3a extraction eliminates source-value leakage entirely.

Solve Results (qwen3.5-flash, n=1, 2 passes, 2 seeds)

Math (20 eval problems)

Config	Seed 42	Seed 43	Mean
Baseline	18/20	20/20	19.0
Concept (v3a)	19/20	19/20	19.0

- Delta: **0.0** — flash is too strong for these 20 problems
- Key validation: **zero concept damage** vs m3's 7 hurts with leaky concepts

LCB (20 eval problems)

Config	Seed 42	Seed 43	Mean
Baseline	17/20	15/20	16.0
Concept (v3a)	20/20	17/20	18.5

- Delta: **+2.5** — consistent improvement across seeds

Retry Recovery: Where Concepts Shine

Domain	Mode	s42 Retry	s43 Retry
LCB	Baseline	2/5 (40%)	3/8 (38%)
LCB	**Concept**	**6/6 (100%)**	3/6 (50%)

Concept hints give the model a **new angle** when its first approach fails.

Seed 42: concept recovered **all 6** failed problems on retry vs baseline's 2/5.

v3a vs v3b: Domain-Specific Tradeoff

v3b drops `implementation` and `parameters` fields (40% shorter hints).

Domain	Variant	s42	s43	Mean
Math	Baseline	18	20	19.0
Math	v3a	19	19	19.0
Math	**v3b**	**20**	**20**	**20.0**
LCB	Baseline	17	15	16.0
LCB	**v3a**	**20**	**17**	**18.5**
LCB	v3b	17	15	16.0

- **Math**: v3b wins — leaner hints let the solver focus
- **LCB**: v3a wins — procedural details matter for algorithm selection
- **v3a is the best general variant** for both domains

Why Implementation Details Matter for Code

Math: "Vieta's Formulas" as a name already tells you what to do.

Code: "Digit DP" as a name is less useful than:

> *"Digit DP with tight bound tracking — iterate over digit positions, maintain (position, tight, accumulated_state) with memoization"*

Code problems need to know **how** to apply an algorithm, not just its name. Dropping implementation details makes v3b = baseline for LCB.

Summary

- **v3a eliminates leakage**: 76% → 0% for math
- **Zero concept damage**: no more misleading hints
- **LCB +2.5/20** net improvement (strongest on retry)
- **v3a is best general variant** across both domains
- Retry recovery is where concepts add the most value

Appendix: Side-by-Side Comparison

Source	v1 (Leaky)	v3a (Clean)
cmath_6446	`"compute 160mi / 5hr = 32mph"`	`"divide total path length by total time"`
cmath_5659	`"base CM = 2, height = 8 → area = 8"`	`"compute sum of cross-products of adjacent coords"`
cmath_2171	`"center at (4,4) for 7×7 grid"`	`"identify orbits of domain under group action"`